Inequalities in one triangle

ANSWER KEY

Core Concepts: Inequalities in one triangle

1 In $\triangle ABC$ studying inequalities in one triangle, $m\angle A = 43^\circ$ and $m\angle B = 45^\circ$. What is $m\angle C$?

a. 92° ✓

b. 102°

c. 84°

d. 137°

Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (43^\circ + 45^\circ) = 92^\circ$.

2 A geometric figure scaled by factor $k = 2$ has an original side length of 7 cm. What is the corresponding image length?

a. 12 cm

b. 14 cm ✓

c. 21 cm

d. 28 cm

Multiplying the original length by the scale factor gives $7 \times 2 = 14$ cm.

3 On the coordinate plane, a line with slope $m = 4$ passes through $(0, 6)$. What is its y-coordinate when $x = 1$?

a. 6

b. 12

c. 10 ✓

d. 14

Substituting $x = 1$ into $y = 4x + 6$ gives $y = 4(1) + 6 = 10$.

Procedural Fluency & Algebraic Manipulation

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4 In $\triangle ABC$ studying inequalities in one triangle, $m\angle A = 43^\circ$ and $m\angle B = 52^\circ$. What is $m\angle C$?

a. 85° ✓

b. 95°

c. 77°

d. 137°

Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (43^\circ + 52^\circ) = 85^\circ$.

5 A geometric figure scaled by factor $k = 3$ has an original side length of 7 cm. What is the corresponding image length?

a. 18 cm

b. 21 cm ✓

c. 28 cm

d. 63 cm

Multiplying the original length by the scale factor gives $7 \times 3 = 21$ cm.

6 On the coordinate plane, a line with slope $m = 4$ passes through $(0, 6)$. What is its y-coordinate when $x = 2$?

a. 10

b. 16

c. 14 ✓

d. 18

Substituting $x = 2$ into $y = 4x + 6$ gives $y = 4(2) + 6 = 14$.

Graphical & Geometric Interpretation

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7 In $\triangle ABC$ studying inequalities in one triangle, $m\angle A = 43^\circ$ and $m\angle B = 59^\circ$. What is $m\angle C$?

a. 78° ✓

b. 88°

c. 70°

d. 137°

Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (43^\circ + 59^\circ) = 78^\circ$.

8 A geometric figure scaled by factor $k = 4$ has an original side length of 7 cm. What is the corresponding image length?

a. 24 cm

b. 28 cm ✓

c. 35 cm

d. 112 cm

Multiplying the original length by the scale factor gives $7 \times 4 = 28$ cm.

9 On the coordinate plane, a line with slope $m = 4$ passes through $(0, 6)$. What is its y-coordinate when $x = 3$?

a. 14

b. 20

c. 18 ✓

d. 22

Substituting $x = 3$ into $y = 4x + 6$ gives $y = 4(3) + 6 = 18$.

Exam-Style Problem Solving & Modeling

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10 In $\triangle ABC$ studying inequalities in one triangle, $m\angle A = 43^\circ$ and $m\angle B = 66^\circ$. What is $m\angle C$?

a. 71° ✓

b. 81°

c. 63°

d. 137°

Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (43^\circ + 66^\circ) = 71^\circ$.

11 A geometric figure scaled by factor $k = 5$ has an original side length of 7 cm. What is the corresponding image length?

a. 30 cm

b. 35 cm ✓

c. 42 cm

d. 175 cm

Multiplying the original length by the scale factor gives $7 \times 5 = 35$ cm.

12 On the coordinate plane, a line with slope $m = 4$ passes through $(0, 6)$. What is its y-coordinate when $x = 4$?

a. 18

b. 24

c. 22 ✓

d. 26

Substituting $x = 4$ into $y = 4x + 6$ gives $y = 4(4) + 6 = 22$.

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